Holevo's Bound and the Holevo Information

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1 Holevo's Bound

In quantum information theory, a central question is how much classical information can be reliably extracted from a quantum system. This question is answered, in part, by *Holevo's bound*, which provides a fundamental upper limit on the amount of accessible classical information that can be obtained from a quantum ensemble.

1.1 Accessible Information

Suppose Alice wishes to send classical information to Bob using quantum states. She prepares quantum states from an ensemble

$$\mathcal{E} = \{(p_i, \rho_i)\}_i \,,$$

where:

- p_i is the probability that Alice prepares the quantum state ρ_i,
- ρ_i is a density operator acting on a finite-dimensional Hilbert space,
- $\sum_i p_i = 1$.

Bob performs a quantum measurement on the received state in order to infer which classical symbol Alice sent. Let X denote the random variable corresponding to Alice's classical input, and let Y denote the random variable corresponding to Bob's measurement outcome.

The *accessible information*, denoted $I_{\text{acc}}(X : Y)$, is defined as the maximum classical mutual information between X and Y, optimized over all possible measurements Bob may perform:

$$I_{\text{acc}}(X : Y) := \max_{\text{measurements}} I(X : Y).$$

1.2 Holevo Information

Holevo's bound states that the accessible information is upper bounded by a quantity known as the *Holevo information* (or Holevo χ quantity).

[Holevo's Bound] For any quantum ensemble $\{(p_i, \rho_i)\}_i$, the accessible information satisfies

$$I_{\text{acc}}(X : Y) \leq \chi, \tag{11.36}$$

where the Holevo information χ is defined as

$$\chi := S(\eta) - \sum_i p_i S(\rho_i). \tag{11.37}$$

Here,

$$\eta := \sum_i p_i \rho_i \tag{11.38}$$

is the *average state* of the ensemble, and $S(\cdot)$ denotes the von Neumann entropy.

1.3 Von Neumann Entropy

The von Neumann entropy of a quantum state ρ is defined as

$$S(\rho) := -\operatorname{Tr}(\rho \log_2 \rho).$$

It is the quantum analogue of the Shannon entropy and quantifies the uncertainty or mixedness of a quantum state.

1.4 Interpretation of Holevo's Bound

The Holevo information χ represents the maximum amount of classical information (in bits) that can be extracted per use of a quantum channel when Alice encodes classical information into quantum states drawn from the ensemble $\{(p_i, \rho_i)\}$.

Important observations:

- If all ρ_i are *orthogonal*, then χ can reach the Shannon entropy of the classical distribution $\{p_i\}$.
- If the states ρ_i are *non-orthogonal*, they cannot be perfectly distinguished, and thus χ is strictly less than the classical entropy.
- If all ρ_i are pure states, then $S(\rho_i) = 0$, and the Holevo information reduces to

$$\chi = S(\eta).$$

Thus, Holevo's bound captures the fundamental limitation imposed by quantum mechanics on the transmission of classical information using quantum states.

2 Conclusion

Holevo's bound is a cornerstone result in quantum information theory. It establishes that no measurement strategy can extract more classical information from a quantum ensemble than what is allowed by the Holevo χ quantity, regardless of the measurement performed. This bound plays a crucial role in quantum communication, quantum cryptography, and the theory of quantum channels.